Complete Math Roadmap

Canadian Curriculum + ML/CS Math — Aligned to Your Coding Timeline

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Big Picture Alignment

Period	Coding Phase	Math Priority
Summer before Gr. 11	Phases 0–1 (DSA, Fundamentals)	MCR3U (Functions) + Linear Algebra start
Gr. 11 school year	Phases 0–2 (DSA, Backend)	MCR3U in school + Stats/Probability evenings
Summer before Gr. 12	Phases 2–4 (Backend, DevOps, System Design)	MHF4U (Advanced Functions) + finish Linear Algebra
Gr. 12 school year	Phases 5–6 (Math for ML, Classical ML)	MHF4U + MCV4U (Calculus) in school
Summer after Gr. 12	Phases 7–8 (Deep Learning, Applied AI)	Full ML math: Calculus, Probability, Information Theory

> **Rule:** Never let math block coding. Learn math *just-in-time* as your coding phases demand it. School carries the curriculum — use your own time for the CS-specific slices school won't teach.

🌞 Summer Before Grade 11

~70 days · 1 hr/day math + your coding hours

Coding focus: finishing CS50P, starting C++, building the DSA habit. Math focus: MCR3U front-loaded so school feels like review, plus starting linear algebra now (no calculus needed).

Block 1 — MCR3U: Functions (Days 1–30)

1 hr/day. Finish before school starts so Gr. 11 math is revision, not new learning.

Days	Topic	Resource	Why It Matters for CS
1–3	Functions, domain/range, notation	Khan Academy — Functions	Function composition mirrors programming functions
4–6	Transformations (shift, reflect, stretch)	Khan Academy — Transformations	Activation function intuition later
7–8	Inverse functions	Khan Academy — Inverse Functions	Critical for understanding bijections in crypto/hashing
9–11	Quadratics: vertex form, completing the square	Khan Academy — Quadratics	Optimization intuition
12–13	Quadratic formula, discriminant	Khan Academy — Quadratics	
14–15	Polynomial behaviour, factor/remainder theorem	Khan Academy — Polynomials	
16–17	Rational functions, asymptotes	Khan Academy — Rational Functions	
18–21	Exponential functions — growth/decay, transformations	Khan Academy — Exponential Functions	Direct prerequisite for learning rate schedules, sigmoid
22–23	Solving exponential equations	Khan Academy — Exponential Equations	
24–26	Trig — unit circle, radians, CAST rule	Khan Academy — Trigonometry	Fourier transforms later; positional encoding in transformers
27–28	Graphing sin/cos/tan, transformations	Khan Academy — Trig Graphs	
29–30	Sine law, cosine law	Khan Academy — Trigonometry	

Block 2 — Sequences & Series (Days 31–40)

Days	Topic	Resource	Why It Matters for CS
31–33	Arithmetic sequences & series	Khan Academy — Sequences	Algorithm complexity intuition
34–36	Geometric sequences & series	Khan Academy — Geometric Sequences	Infinite series, gradient descent convergence
37–38	Infinite geometric series	Khan Academy — Series	Neural net weight decay intuition
39–40	Pascal's triangle, intro to combinatorics	Khan Academy — Combinatorics	Probability foundation

Block 3 — Linear Algebra Start (Days 41–65)

This is Phase 5 of your coding roadmap pulled forward. No calculus needed.

Days	Topic	Resource	How to Learn
41–43	Vectors — addition, scalar mult., dot product	3Blue1Brown "Essence of Linear Algebra" Ep. 1–2	Watch → implement in NumPy same day
44–46	Linear transformations, matrix multiplication	3B1B Ep. 3–4	Watch → build matrix multiply by hand, verify with `np.dot`
47–49	Determinants	3B1B Ep. 5–7	Watch → implement
50–52	Systems of equations, Gaussian elimination	Khan Academy Linear Algebra	Read → solve a 3×3 system in pure Python
53–55	Matrix inverse, rank, null space	3B1B + Khan Academy	Watch/read → implement
56–58	Eigenvectors & eigenvalues	3B1B Ep. 14	Watch → find eigenvalues of a 2×2 by hand, verify with `np.linalg.eig`
59–61	PCA intuition (eigenvectors applied)	StatQuest "PCA" video	Watch → PCA on a toy 2D dataset in NumPy
62–64	Matrix norms, orthogonality, projections	Khan Academy	Read → implement
65	Review & consolidation	—	Build a mini linear algebra library in Python from scratch

Block 4 — Review & Buffer (Days 66–70)

Catch-up days. Re-solve anything shaky. No new material.

📚 Grade 11 School Year

~10 months · ~30–45 min/day on math outside school

School handles MCR3U. Evening time goes to **probability & statistics** — the other half of Phase 5 that school won't cover adequately.

Probability & Statistics (evenings, alongside school)

~3 sessions/week, 30–45 min. Roughly October → March.

Week	Topic	Resource	How to Learn
1–2	Counting: permutations, combinations	Khan Academy Stats	Timed reps on exercises
3–4	Basic probability: sample spaces, events, complement rule	StatQuest + Khan Academy	Watch → simulate in Python (flip coins, roll dice)
5–6	Conditional probability, independence	StatQuest	Watch → verify with Python simulations
7–8	Bayes' theorem	StatQuest "Bayes' theorem"	Watch → write a Naive Bayes classifier from scratch
9–10	Discrete distributions: Bernoulli, Binomial, Poisson	StatQuest	Watch → simulate each distribution in NumPy
11–12	Continuous distributions: Uniform, Normal (Gaussian)	StatQuest	Watch → sample, plot histograms, compute z-scores
13–14	Expectation, variance, standard deviation	Khan Academy	Read → implement by hand on small datasets
15–16	Central Limit Theorem	StatQuest	Watch → simulate: show CLT empirically in Python
17–18	Covariance, correlation	StatQuest	Watch → correlation matrix in NumPy
19–20	Maximum Likelihood Estimation (MLE)	StatQuest	Watch → MLE for a Gaussian by hand
21–22	Hypothesis testing, p-values (conceptual)	StatQuest	Watch → implement a t-test manually
23–24	Information theory basics: entropy, cross-entropy, KL divergence	Colah's blog (blog posts)	Read → compute cross-entropy loss manually

> **Why this order?** By the time you hit Phase 6 (Classical ML) in coding, you'll already understand *why* cross-entropy is the loss function and what MLE is actually doing.

🌞 Summer Before Grade 12

~70 days · 1 hr/day

Coding focus: wrapping up Backend (Phase 2), DevOps (Phase 4), System Design (Phase 3). Math focus: MHF4U preview + calculus entry point.

Block 1 — MHF4U: Advanced Functions Preview (Days 1–25)

Front-load so Gr. 12 school math is easy.

Days	Topic
1–3	Polynomial functions — graphs, end behaviour, multiplicity of roots
4–6	Dividing polynomials, factor theorem review
7–9	Rational functions — intercepts, asymptotes (vertical, horizontal, oblique)
10–12	Logarithms — definition, laws, change of base
13–15	Solving log equations, graphing log functions
16–18	Exponential & log relationship (inverses)
19–21	Trig identities — Pythagorean, sum/difference formulas
22–24	Solving trig equations, general solutions
25	Review

Block 2 — Calculus Entry Point (Days 26–55)

MCV4U preview. Chain rule is what you need for backprop — go deep on it.

Days	Topic	Why It Matters for CS
26–28	Limits — intuition, limit laws	Foundation for derivatives
29–32	Derivatives from first principles	Understanding what a gradient *is*
33–35	Differentiation rules: power, product, quotient	
36–39	Chain rule	This IS backpropagation. Go slow. Master it.
40–42	Derivatives of exponentials & logs (e^x, ln x)	Softmax, log-loss derivatives
43–45	Derivatives of trig functions	
46–48	Partial derivatives — intro	Direct prerequisite for gradient descent in ML
49–51	Gradients & gradient vectors	The math behind every optimizer (SGD, Adam)
52–54	Optimization — critical points, 2nd derivative test	Loss minimization intuition
55	Review	

Block 3 — Linear Algebra Completion + Multivariable (Days 56–70)

Days	Topic	Why It Matters for CS
56–58	Jacobian matrix intuition	Backprop through layers
59–61	Gradient descent implemented in NumPy	Linear regression with gradient descent from scratch — no sklearn
62–64	SVD intuition (Singular Value Decomposition)	Used in recommendation systems, compression
65–67	Vector calculus: gradient, divergence (light)	Needed for understanding loss landscape

Days	Topic	Why It Matters for CS
68–70	Backpropagation from scratch — derive it mathematically	Implement a 2-layer net backward pass in pure NumPy

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📚 Grade 12 School Year

~10 months

School handles MHF4U (Advanced Functions) and MCV4U (Calculus & Vectors). Evening time goes to solidifying ML math and beginning deeper CS math.

Evening Math: ML Math Consolidation + CS Theory

~3 sessions/week, 30–45 min.

Period	Topic	Resource
Sept–Oct	Multivariate calculus — chain rule in matrix form, Jacobians	*Mathematics for Machine Learning* (MML book, free)
Nov–Dec	Optimization theory — convexity, saddle points, Adam/momentum math	MML book Ch. 7
Jan–Feb	Linear algebra deep dive — SVD, matrix factorization, spectral methods	MML book Ch. 2–4
Mar–Apr	Probability theory formal — measure theory lite, conjugate priors, distributions in depth	MML book Ch. 6
May–Jun	Information theory — entropy, mutual information, KL divergence applications	*Elements of Information Theory* Ch. 1–2

🌞 Summer After Grade 12

~70 days — aligns with Phases 7–8: Deep Learning + Applied AI

This is where the math pays off: implementing backprop, attention mechanisms, and loss functions for real.

Days	Topic	Coding Tie-in
1–5	Backprop derivation — scalar → vector → matrix form	Phase 7: implement in pure NumPy
6–10	Loss functions derived — MSE, cross-entropy, KL divergence	Why these functions, not others
11–15	Activation functions — sigmoid, tanh, ReLU, derivatives	Vanishing gradient problem — understand it mathematically
16–20	Attention mechanism math — scaled dot-product, softmax derivation	Phase 7: implement attention from scratch in PyTorch
21–25	Convolutional math — cross-correlation, filter math	CNN intuition
26–30	Regularization math — L1/L2 as priors (Bayesian view), dropout	

Block 2 — Advanced CS Math (Days 31–55)

Days	Topic	Why It Matters
31–35	Combinatorics & counting — generating functions, recurrences	DSA: DP state counting, CCC problems
36–40	Graph theory — trees, cycles, planarity, graph representations	DSA: Phase 0 Gold topics
41–45	Discrete math — logic, sets, relations, proofs	CS fundamentals, algorithm correctness
46–50	Number theory — modular arithmetic, Fermat's little theorem, RSA	Cryptography in backend/security
51–55	Big-O formal — asymptotic notation proofs, recurrence relations (Master theorem)	Analyzing your own algorithms rigorously

Days	Topic
56–59	Bayesian ML — Bayesian linear regression, MAP estimation
60–63	Statistical learning theory — bias-variance tradeoff formally, VC dimension (intro)
64–67	Embedding math — vector spaces, cosine similarity derivation, why dot product works
68–70	Review + fill gaps identified from Phase 7–8 coding work

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Complete Curriculum Checklist

Ontario School Curriculum (all covered)

- **MPM1D/2D** (Gr. 9–10) — assumed done
- **MCR3U** (Gr. 11 Functions) — Summer before Gr. 11 + school year
- **MHF4U** (Gr. 12 Advanced Functions) — Summer before Gr. 12 + school year
- **MCV4U** (Gr. 12 Calculus & Vectors) — Summer before Gr. 12 + school year
- **MDM4U** (Data Management — optional) — covered by stats self-study, better than the course

ML/CS Math (all covered)

- Linear Algebra — Summer Gr. 11, finished Gr. 12
- Probability & Statistics — Gr. 11 evenings
- Calculus (derivatives, chain rule, partial derivatives) — Summer Gr. 12
- Multivariable Calculus / Gradient Descent math — Summer Gr. 12
- Information Theory — Gr. 12 evenings
- Backprop math — Summer after Gr. 12
- Attention math — Summer after Gr. 12
- Discrete Math / Combinatorics — Summer after Gr. 12
- Graph Theory — Summer after Gr. 12
- Number Theory — Summer after Gr. 12

Golden Rules

1. **School carries the curriculum.** Never spend self-study time re-doing what school covers adequately. Use your hours for the CS math school skips.
2. **Linear algebra before calculus.** You can start linear algebra now — no calculus needed. It pays off in Phase 6 before you even touch derivatives.
3. **Simulate, don't just calculate.** Every stats and probability concept should be verified with a Python simulation the same day. `numpy.random` is your proof-checker.
4. **The chain rule is backprop.** Spend more time here than anything else in calculus. Master it symbolically, then implement it in code.
5. **MML book is your capstone math text.** *Mathematics for Machine Learning* (free at mml-book.github.io) ties all of this together. Read it in Gr. 12 once school has given you the calculus foundation.